Fredrik Pettit

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Education

University of British Columbia — Vancouver, BC
B.Sc. Computer Science, Artificial Intelligence Option — *Expected Graduation: May 2028*

Experience

AI Engineer Intern — May 2026 – Sep 2026
AEM — *Hayward, CA*

- Saved **$16K** a year in staff time with an AI support agent that answers customers in English and German
- Deployed the agent as a FastAPI service on Azure Kubernetes that streams each reply to the customer within 3 seconds and hands anything it can't answer to the help desk through a webhook
- Raised auto-resolution from 32% to **43%** with a RAG pipeline on Azure using hybrid search over 4,000 pages of company docs after tracing most escalations to German and acronym-heavy customer questions
- Caught a 6-point drop in correct answers from a silent model update with an automated 300-case regression eval
- Cut monthly budget reporting from **6+ hours to 5 minutes** with a LangChain agent that turns NetSuite exports into Excel reports through Microsoft Graph API and sends unsure expenses to finance for human review

Software Engineering Intern — Jan 2026 – Apr 2026
ELLUSTROS — *The Blue Mountains, ON*

- Built the FastAPI backend that recommends best-fit chairs to 300+ users by scoring each chair's ergonomic attributes against their body measurements from the React Native app
- Shipped PostgreSQL schema changes through Alembic without losing user data or breaking older app versions
- Cut failed photo uploads on slow mobile networks by **70%** with presigned URLs that send images straight to S3
- Kept the app and API in sync by testing requests against Pydantic models in GitHub Actions CI/CD

Teaching Assistant | CPSC 310, CPSC 110 — Aug 2025 – Present
UBC Department of Computer Science — *Vancouver, BC*

- Earned a 4.9/5 average student rating while teaching 900+ students in UBC's largest computer science course
- Wrote 250+ randomized JavaScript questions and Python autograders for CPSC 310

Projects

Email Synced Cashflow Tracker | *Java, Spring Boot, Go, React, PostgreSQL, Redis, Pub/Sub, Docker*

- Built a Spring Boot and React app that tracks orders and profit for a personal resale business with $6K+ in sales
- Automated order entry with a concurrent Go worker that turns eBay, Depop and Grailed sale emails from Gmail through Pub/Sub into orders while skipping duplicates by order ID
- Sped up the sales dashboard API **36×** by caching report queries in Redis and clearing them on each new order

YouTube Shorts AI Optimizer | *Python, FastAPI, Claude API, Gemini API, Google Cloud Run, BigQuery*

- Built an AI agent with **100+** users that reviews draft YouTube Shorts and grew a personal channel to **45M** views
- Completed **99.7%** of 1,300 reviews exactly once with Cloud Tasks retries and Cloud Run jobs keyed by video hash
- Cut p95 latency from 95s to 18s by tracing agent steps with OpenTelemetry and parallelizing Claude tool calls

Apex FSD - StormHacks 2026 (Winner) | *Python, PyTorch, OpenCV, PySDL2*

- Won Best Python Hack at Western Canada's largest hackathon with a fully autonomous driver that plays Forza Horizon by turning a physical wheel in a 100 Hz PySDL2 loop and hands control back once a human grabs it

Scheming - HackMIT 2026 | *Python, Claude API, OpenAI API, Modal, MongoDB*

- Built a lie detector for AI agents playing Werewolf that flags the exact line where a player lies, and designed a MongoDB schema with one collection per teammate so a team of 4 could build in parallel without conflicts

Technical Skills

Languages & Databases: Java, Python, Go, TypeScript/JavaScript, C++, SQL, PostgreSQL, Redis, MongoDB
Frameworks: Spring Boot, FastAPI, Node.js, React Native, SQLAlchemy, Pydantic, pandas, NumPy, pytest, Jest
Cloud & DevOps: AWS (S3), GCP (Cloud Run, Pub/Sub), Azure, Kubernetes, Docker, Linux, GitHub Actions
AI/ML: PyTorch, TensorFlow, scikit-learn, OpenCV, LangChain, LangGraph, MCP, OpenTelemetry, guardrails
Concepts: Data Structures & Algorithms, Object-Oriented Design, Scalable Systems, Back-End Development, RESTful APIs, SQL & NoSQL Databases, Performance Optimization, CI/CD, Machine Learning, Agentic AI, Model Evaluation